

Gebze Technical University
Department of Mechanical Engineering

Recitation 3

Undamped Free Vibration

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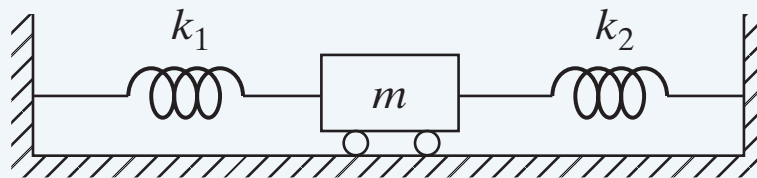
Self-study guide. These 32 questions develop the Week 3 topics through explanations and worked calculations. Each question is followed immediately by its solution.

Levels: Foundation builds the basic relations; Application uses them in a specified model; Challenge combines several steps or requires closer interpretation. The questions progress within each topic. Angles are in radians unless degrees are stated. Use $g = 9.81 \text{ m/s}^2$.

Question 1 | Four routes to one equation

Foundation

A mass m moves horizontally without friction between two linear springs k_1 and k_2 . Derive its equation of motion using (a) Newton's law, (b) D'Alembert's principle, (c) virtual work, and (d) conservation of energy.



Take x positive to the right from equilibrium.

Solution 1

1. Newton's law

For a positive displacement x , the left spring stretches and the right spring compresses. Both incremental forces act leftward:

$$\sum F_x = -k_1x - k_2x = m\ddot{x}.$$

Thus $m\ddot{x} + (k_1 + k_2)x = 0$. Vertical weight and support reactions balance.

2. D'Alembert's principle

Add the inertia force $-m\ddot{x}$ to the actual forces:

$$-k_1x - k_2x - m\ddot{x} = 0.$$

The inertia force opposes acceleration; it does not necessarily oppose velocity.

3. Virtual work

Choose any small, constraint-compatible horizontal displacement δx at a fixed instant. The normal reaction and weight do no virtual work. Dynamic virtual work gives

$$[-(k_1 + k_2)x - m\ddot{x}] \delta x = 0.$$

Because δx is arbitrary, the bracket must vanish. Virtual displacement is not $\dot{x} dt$; time is held fixed.

Solution 1 (continued)**4. Conservation of energy**

$$T = \frac{1}{2}m\dot{x}^2, \quad U = \frac{1}{2}(k_1 + k_2)x^2.$$

Differentiate the constant total energy:

$$\frac{d}{dt}(T + U) = \dot{x} [m\ddot{x} + (k_1 + k_2)x] = 0.$$

For the moving portions of the motion, the bracket is zero; the same equation holds at isolated turning points by continuity. Hence all four methods give

$$m\ddot{x} + (k_1 + k_2)x = 0, \quad \omega_n = \sqrt{\frac{k_1 + k_2}{m}}.$$

Question 2 | When can energy determine frequency?**Foundation**

Can conservation of mechanical energy be used to obtain the equation of motion of every single-degree-of-freedom system? State the assumptions needed for Rayleigh's energy method and explain why maximum kinetic and potential energies can be equated.

Solution 2**1. Identify when mechanical energy is conserved**

For a conservative system with no external energy input, $T + U = E$ is constant. Ideal fixed constraints do no work. Damping, friction, or an applied time-dependent force generally gives $dE/dt \neq 0$; simple conservation cannot then be used without accounting for that work.

2. Express both energies using one coordinate

For small oscillations about a stable equilibrium of a linear system,

$$T = \frac{1}{2}m_{\text{eq}}\dot{q}^2, \quad U = \frac{1}{2}k_{\text{eq}}q^2.$$

Both equivalent quantities must refer to the *same* coordinate q . For rotation, they become mass moment of inertia and torsional stiffness.

3. Compare two different instants

Let $q = Q \cos(\omega_n t)$. At a displacement extremum, $\dot{q} = 0$, so $E = U_{\text{max}} = \frac{1}{2}k_{\text{eq}}Q^2$. At equilibrium passage, $q = 0$, so $E = T_{\text{max}} = \frac{1}{2}m_{\text{eq}}\omega_n^2 Q^2$. Therefore

$$T_{\text{max}} = U_{\text{max}} \implies \boxed{\omega_n^2 = \frac{k_{\text{eq}}}{m_{\text{eq}}}}.$$

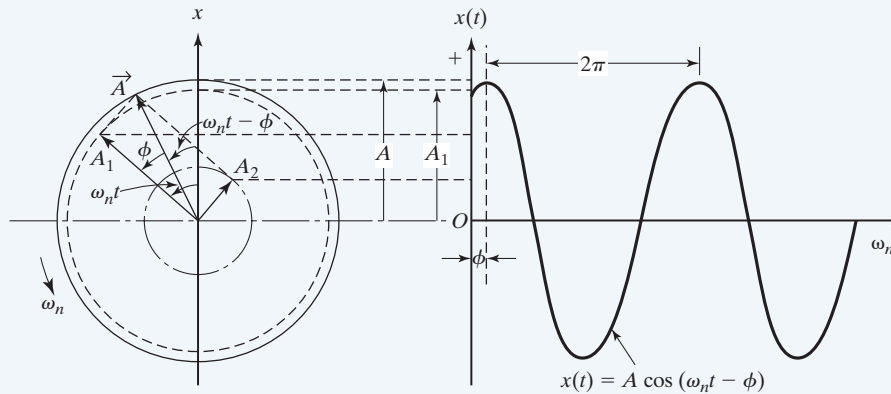
The maxima occur at different times. Their equality does not require resonance or an applied periodic force.

4. Recognize the approximation for a flexible body

For a beam or other distributed system, assume a displacement shape $u(s, t) = \psi(s)q(t)$ that satisfies the essential support constraints. Compute strain and kinetic energies from that same shape. The resulting frequency is an estimate because the assumed shape may differ from the true mode. A poor shape or omitted inertia can give a poor estimate even if the algebra is correct.

Question 3 | Rotating amplitude vectors**Foundation**

Using the rotating-vector construction, explain the response for $x_0 > 0$ and $v_0 > 0$. Identify the two component amplitudes, obtain A and ϕ , and locate the first displacement maximum. The vertical axis in this construction represents displacement.



Rotating amplitude vectors and their vertical projection.

Solution 3**1. Determine the two components**

The solution and its derivative are

$$x = A_1 \cos \omega_n t + A_2 \sin \omega_n t, \quad \dot{x} = -\omega_n A_1 \sin \omega_n t + \omega_n A_2 \cos \omega_n t.$$

At $t = 0$, $A_1 = x_0$ and $A_2 = v_0/\omega_n$.

2. Interpret the vectors

Initially, $\vec{A}_1$ points upward and $\vec{A}_2$ points rightward. They are perpendicular and rotate counterclockwise at angular speed ω_n . Their vertical projections are respectively $A_1 \cos \omega_n t$ and $A_2 \sin \omega_n t$. The vertical projection of their sum is $x(t)$.

3. Combine the components

Expand $A \cos(\omega_n t - \phi)$ and compare coefficients:

$$A \cos \phi = x_0, \quad A \sin \phi = \frac{v_0}{\omega_n}.$$

Thus

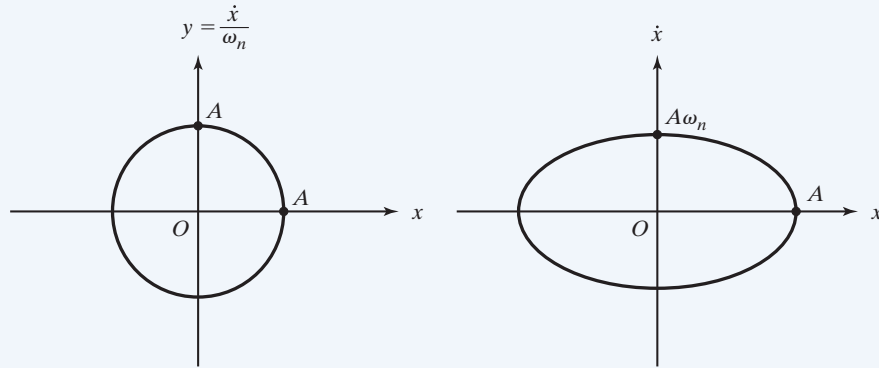
$$A = \sqrt{x_0^2 + (v_0/\omega_n)^2}, \quad \phi = \tan^{-1} \frac{v_0}{\omega_n x_0} \quad (0 < \phi < \pi/2).$$

4. Check the motion

The resultant starts to the right of the upward axis and reaches that axis after rotating through ϕ . Consequently $t_{\text{peak}} = \phi/\omega_n$, $x(t_{\text{peak}}) = A$, and $\dot{x}(t_{\text{peak}}) = 0$. At $t = 0$, $\dot{x} = A\omega_n \sin \phi > 0$, as required. This figure is not an x -velocity phase plane.

Question 4 | Phase plane and energy**Foundation**

For $m\ddot{x} + kx = 0$, eliminate time from the displacement and velocity responses. Interpret the curves in the $(x, \dot{x}/\omega_n)$ and $(x, \dot{x})$ planes, determine the direction of motion, and relate orbit size to total energy.



Scaled and unscaled phase-plane coordinates.

Solution 4**1. Eliminate time**

Put $\alpha = \omega_n t - \phi$. Then $x = A \cos \alpha$ and $\dot{x} = -A\omega_n \sin \alpha$. Squaring and adding gives

$$x^2 + \left(\frac{\dot{x}}{\omega_n}\right)^2 = A^2.$$

Equal scales on both axes produce a circle of radius A in the scaled plane. In the unscaled plane,

$$\frac{x^2}{A^2} + \frac{\dot{x}^2}{A^2\omega_n^2} = 1,$$

an ellipse with intercepts $\pm A$ and $\pm A\omega_n$.

2. Determine the direction

At the rightmost point $(A, 0)$, acceleration is $\ddot{x} = -\omega_n^2 A < 0$. Velocity immediately becomes negative, moving the point downward. The orbit is therefore traversed clockwise. At the top point, $x = 0$ and $\dot{x} > 0$, so the point moves rightward, confirming the direction.

3. Relate the curve to energy

Since $k = m\omega_n^2$,

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = \frac{1}{2}k \left[x^2 + (\dot{x}/\omega_n)^2 \right] = \boxed{\frac{1}{2}kA^2}.$$

The orbit is a constant-energy curve. Doubling A quadruples E . At $x = \pm A$, all energy is potential; at $x = 0$, it is kinetic. The orbit does not spiral inward because there is no damping.

Question 5 | Frequency from static compression**Foundation**

An industrial press rests on a rubber pad that compresses 5 mm under the press's weight. Treat the pad as a linear spring and determine the natural frequency.

Solution 5**1. Use the static measurement**

At equilibrium, $k\delta = mg$, so $k/m = g/\delta$. The mass and stiffness need not be known separately.

2. Calculate in SI units

$$\delta = 0.005 \text{ m}, \quad \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{9.81}{0.005}} = \boxed{44.29 \text{ rad/s}}.$$

$$\boxed{f_n = \frac{44.29}{2\pi} = 7.050 \text{ Hz}}, \quad T_n = \frac{2\pi}{44.29} = 0.1419 \text{ s}.$$

3. State the modeling limitation

This result assumes the vibration stiffness equals the linear stiffness inferred from static compression, and the pad's moving mass is negligible. For nonlinear rubber, small vibration uses the tangent stiffness at the loaded equilibrium; a single static compression measurement would then be insufficient.

Question 6 | Changing stiffness**Foundation**

A spring-mass system has a natural period of 0.21 s. Find the new period when its spring stiffness is (a) increased by 50% and (b) decreased by 50%. Keep the mass fixed.

Solution 6**1. Form a ratio**

Using $T_n = 2\pi\sqrt{m/k}$,

$$\frac{T'_n}{T_n} = \sqrt{\frac{k}{k'}}.$$

2. Increase the stiffness

Here $k' = 1.5k$, so

$$T'_n = \frac{0.21}{\sqrt{1.5}} = \boxed{0.1715 \text{ s}}.$$

The stiffer system oscillates faster and has a shorter period.

3. Decrease the stiffness

Here $k' = 0.5k$, so

$$T'_n = \frac{0.21}{\sqrt{0.5}} = \boxed{0.2970 \text{ s}}.$$

The softer system oscillates more slowly. A 50% stiffness change does not produce a 50% period change, because period depends on $k^{-1/2}$.

Question 7 | Identify mass and stiffness**Application**

A spring–mass system has natural frequency 10 Hz. Reducing its stiffness by 800 N/m changes the frequency by 45%. Determine the original mass and stiffness.

Solution 7**1. Choose the physically possible frequency change**

With unchanged positive mass, reducing stiffness reduces frequency. Thus

$$f'_n = 0.55(10) = 5.5 \text{ Hz.}$$

2. Use the squared frequency ratio

$$\left(\frac{5.5}{10}\right)^2 = \frac{k - 800}{k} \implies 0.3025k = k - 800.$$

Therefore

$$k = \frac{800}{0.6975} = 1146.95 \text{ N/m.}$$

3. Find the mass

Convert cyclic frequency to angular frequency before using $k/m = \omega_n^2$:

$$\omega_n = 2\pi(10) = 62.8319 \text{ rad/s,} \quad m = \frac{1146.95}{(62.8319)^2} = 0.29053 \text{ kg.}$$

4. Check the modified system

The new stiffness is 346.95 N/m > 0. Substitution gives

$$\frac{1}{2\pi} \sqrt{\frac{346.95}{0.29053}} = 5.50 \text{ Hz,}$$

which agrees with the specified reduction.

Question 8 | A mass at the middle of a spring**Application**

A uniform helical spring extends 10 mm under a 100 N load. Both ends are then fixed, one above the other, and a 10 kg mass is attached at its midpoint. Neglect spring mass and find the period of small vertical vibration.

Solution 8**1. Find the full spring stiffness**

$$k = \frac{100}{0.010} = 10,000 \text{ N/m.}$$

2. Find the stiffness of each half

For a uniform helical spring, stiffness is inversely proportional to the number of active coils. Each half has half as many coils, hence $k_{\text{half}} = 2k = 20,000 \text{ N/m}$.

3. Use the mass displacement coordinate

For a downward displacement x from equilibrium, the upper half extends by x and the lower half shortens by x . Their incremental forces both restore the mass upward, so

$$k_{\text{eq}} = k_{\text{upper}} + k_{\text{lower}} = 4k = 40,000 \text{ N/m.}$$

The two halves act in parallel for this coordinate, even though they lie on opposite sides of the mass.

4. Compute the period

$$\omega_n = \sqrt{\frac{40,000}{10}} = 63.246 \text{ rad/s,} \quad T_n = \frac{2\pi}{63.246} = 0.09935 \text{ s.}$$

Gravity changes the equilibrium location and cancels from the incremental equation of motion.

Question 9 | Identify a system by adding mass**Application**

A spring–mass system oscillates at 2 Hz. Adding 1 kg to its mass reduces its natural frequency to 1 Hz. Find its original mass and spring stiffness.

Solution 9**1. Write the two measurements**

$$k = m(2\pi \cdot 2)^2 = (m + 1)(2\pi \cdot 1)^2.$$

The spring is unchanged; the added mass contributes inertia only.

2. Eliminate stiffness

Dividing by $(2\pi)^2$ gives $4m = m + 1$. Therefore

$$m = \frac{1}{3} \text{ kg} = 0.3333 \text{ kg}.$$

3. Find stiffness and check

$$k = \frac{1}{3}(4\pi)^2 = 52.638 \text{ N/m}.$$

With the added mass, $\omega'_n = \sqrt{52.638/(4/3)} = 2\pi \text{ rad/s}$, hence $f'_n = 1 \text{ Hz}$. Multiplying mass by four has halved the natural frequency, as expected from $f_n \propto m^{-1/2}$.

Question 10 | Peak measurements and phase ambiguity**Application**

An undamped oscillator has maximum speed 10 cm/s and period 2 s. At $t = 0$, its displacement is 2 cm. Find the amplitude, initial velocity, maximum acceleration magnitude, and phase. The initial direction of motion is not specified.

Solution 10**1. Find frequency and amplitude**

$$\omega_n = \frac{2\pi}{2} = \pi \text{ rad/s}, \quad A = \frac{v_{\max}}{\omega_n} = \frac{0.10}{\pi} = \boxed{0.031831 \text{ m}}.$$

2. Find both possible initial velocities

From $x_0^2 + (v_0/\omega_n)^2 = A^2$,

$$v_0 = \pm \sqrt{v_{\max}^2 - \omega_n^2 x_0^2} = \pm \sqrt{0.10^2 - (\pi \cdot 0.020)^2} = \boxed{\pm 0.077796 \text{ m/s}}.$$

3. Find peak acceleration

Since $\ddot{x} = -\omega_n^2 x$,

$$\boxed{|\ddot{x}|_{\max} = \omega_n^2 A = \omega_n v_{\max} = 0.31416 \text{ m/s}^2}.$$

4. Select phase using the velocity sign

$$\cos \phi = \frac{x_0}{A} = 0.628319.$$

For $v_0 > 0$, $\phi = +0.891406$ rad; for $v_0 < 0$, $\phi = -0.891406$ rad. The two possible responses are

$$\boxed{x(t) = 0.031831 \cos(\pi t \mp 0.891406) \text{ m}}.$$

The upper sign corresponds to positive initial velocity. “Released” cannot be interpreted as “released from rest” here: rest at $x_0 = 0.020$ m would give $v_{\max} = \pi(0.020) = 0.06283$ m/s, inconsistent with the data.

Question 11 | Four initial-condition quadrants**Application**

For $m = 5$ kg and $k = 2000$ N/m, determine the complete free response for each pair:

Case	x_0 (mm)	v_0 (mm/s)
(a)	20	200
(b)	-20	200
(c)	20	-200
(d)	-20	-200

Use $x = A \cos(\omega_n t - \phi)$ and keep the correct phase quadrant.

Solution 11**1. Frequency and common amplitude**

$$\omega_n = \sqrt{2000/5} = 20 \text{ rad/s}, \quad |v_0|/\omega_n = 200/20 = 10 \text{ mm},$$

$$A = \sqrt{20^2 + 10^2} = 22.3607 \text{ mm}.$$

Every case has this amplitude because the energy depends on squares of the initial data.

2. Resolve all four quadrants

Let $\alpha = \tan^{-1}(10/20) = 0.463648$ rad. With $\phi = \text{atan2}(v_0/\omega_n, x_0)$,

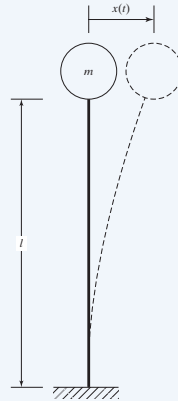
Case	ϕ (rad)	$x(t)$ (mm)
(a)	α	$22.3607 \cos(20t - 0.463648)$
(b)	$\pi - \alpha$	$22.3607 \cos(20t - 2.677945)$
(c)	$-\alpha$	$22.3607 \cos(20t + 0.463648)$
(d)	$-\pi + \alpha$	$22.3607 \cos(20t + 2.677945)$

3. Check every sign directly

At $t = 0$, $x(0) = A \cos \phi$ and $\dot{x}(0) = A\omega_n \sin \phi$. Quadrants I, II, IV, and III respectively reproduce $(+, +)$, $(-, +)$, $(+, -)$, and $(-, -)$. A one-argument inverse tangent would confuse cases (a)/(d) and (b)/(c).

Question 12 | Elevated water tank**Application**

A full water tank has mass 275,000 kg. Its 100 m concrete column has outer diameter 3 m, inner diameter 2.5 m, and $E = 30$ GPa. Neglect column mass. Find the transverse natural frequency and period, the response after release from rest at $x_0 = 0.25$ m, and peak speed and acceleration.



Cantilever-column model; tank treated as a point mass.

Solution 12**1. Obtain the bending stiffness**

For a cantilever tip force P , $\delta = PL^3/(3EI)$, hence

$$I = \frac{\pi}{64}(3^4 - 2.5^4) = 2.058602 \text{ m}^4,$$

$$k = \frac{3EI}{L^3} = \frac{3(30 \times 10^9)(2.058602)}{100^3} = 185,274 \text{ N/m}.$$

2. Frequency and period

$$\omega_n = \sqrt{185,274/275,000} = 0.820807 \text{ rad/s},$$

$$f_n = 0.13064 \text{ Hz}, \quad T_n = 7.6549 \text{ s}.$$

3. Apply the release conditions

Since $v_0 = 0$ and $x_0 > 0$, $A = x_0$ and $\phi = 0$. Therefore

$$x(t) = 0.25 \cos(0.820807t) \text{ m}.$$

4. Differentiate for the peak magnitudes

$$v_{\max} = A\omega_n = 0.20520 \text{ m/s}, \quad |\ddot{x}|_{\max} = A\omega_n^2 = 0.16843 \text{ m/s}^2.$$

This model neglects column inertia, water sloshing, and geometric-stiffness effects. Its stiffness is for bending, not axial compression.

Question 13 | Impact on a beam-supported mass**Challenge**

A mass M is initially at rest at the free end of a massless cantilever of length L and rigidity EI . A second mass m falls through height h onto M and sticks. Determine the subsequent transverse vibration. Assume a short impact, negligible support impulse during impact, and small elastic beam deflection.

Solution 13**1. Impact speed and common velocity**

The falling mass reaches $v_- = \sqrt{2gh}$. Momentum conservation during the short collision gives

$$mv_- = (M + m)v_+, \quad v_+ = \frac{m}{M + m}\sqrt{2gh}.$$

Kinetic energy is not conserved during this perfectly inelastic collision.

2. Define the displacement carefully

Let z be the downward deflection from the unloaded beam position and $k = 3EI/L^3$. Before impact, $z = Mg/k$. After impact, the new equilibrium is $z_{eq} = (M + m)g/k$.

Use $x = z - z_{eq}$ for subsequent vibration. Position is continuous through impact, so

$$x_0 = -\frac{mg}{k}, \quad v_0 = v_+ > 0.$$

The mass starts *above* its new equilibrium and moves downward.

3. Write and solve the post-impact equation

From $(M + m)\ddot{z} = (M + m)g - kz$,

$$(M + m)\ddot{x} + kx = 0, \quad \omega_n = \sqrt{\frac{k}{M + m}}.$$

Therefore the complete response is

$$x(t) = -\frac{mg}{k} \cos \omega_n t + \frac{m\sqrt{2gh}}{(M + m)\omega_n} \sin \omega_n t.$$

The absolute deflection is $z(t) = (M + m)g/k + x(t)$.

4. Amplitude, phase, and checks

$$A = \sqrt{\left(\frac{mg}{k}\right)^2 + \frac{2ghm^2}{k(M + m)}}, \quad \phi = \text{atan2}\left(\frac{v_+}{\omega_n}, -\frac{mg}{k}\right).$$

Here $\pi/2 < \phi < \pi$ for $h > 0$. Substitution at $t = 0$ recovers both initial conditions. Using only the impact velocity would miss the additional response caused by the equilibrium shift.

Question 14 | Young's modulus from a vibration test**Application**

A beam is **fixed at both ends**, has length $L = 1$ m and square cross section $5 \text{ mm} \times 5 \text{ mm}$, and carries 2.3 kg at midspan. Its transverse natural frequency is 30 rad/s . Neglect beam mass and determine E .

Solution 14**1. Find the correct center stiffness**

For a central downward force P , symmetry gives reactions $P/2$ and end moments $-PL/8$. On the left half,

$$EIy''(s) = -\frac{PL}{8} + \frac{P}{2}s.$$

With the signed deflection positive upward, the clamped conditions $y(0) = y'(0) = 0$ give

$$EIy(s) = -\frac{PLs^2}{16} + \frac{Ps^3}{12}.$$

At $s = L/2$, $y = -PL^3/(192EI)$. Thus the *magnitude* of the downward deflection is $\delta = PL^3/(192EI)$, giving

$$k = \frac{P}{\delta} = \frac{192EI}{L^3}.$$

2. Compute the area moment of inertia

$$I = \frac{b^4}{12} = \frac{(0.005)^4}{12} = 5.208333 \times 10^{-11} \text{ m}^4.$$

3. Use the frequency measurement

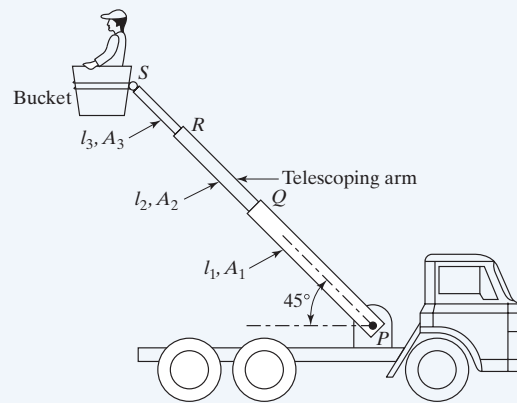
Since $k = m\omega_n^2$,

$$E = \frac{m\omega_n^2 L^3}{192I} = \frac{2.3(30)^2(1)^3}{192(5.208333 \times 10^{-11})} = \boxed{207 \text{ GPa}}.$$

The required stiffness is 2070 N/m . A simply supported beam would have $k = 48EI/L^3$ and give a different answer; support conditions are essential data.

Question 15 | Axial model of a telescoping boom**Application**

The cockpit and firefighter at the end of a 45° boom weigh 2000 N. Each of its three segments is 3 m long; their areas are 20, 10, and 5 cm^2 , and $E = 2.1 \times 10^{11} \text{ Pa}$. Use the idealized **vertical-only, axial-deformation** model, neglect boom mass and bending, and find the vertical natural frequency.



Telescoping firetruck boom.

Solution 15**1. Combine axial compliances**

The segment stiffnesses $k_i = EA_i/L_i$ are 1.4×10^8 , 7.0×10^7 , and $3.5 \times 10^7 \text{ N/m}$. They carry the same axial force, so

$$\frac{1}{k_b} = \sum_i \frac{L_i}{EA_i} = 5.0 \times 10^{-8} \text{ m/N}, \quad k_b = 2.0 \times 10^7 \text{ N/m}.$$

2. Transform to the vertical coordinate

For a constrained vertical displacement x , the axial extension is $x \sin 45^\circ$. Equating strain energies gives

$$\frac{1}{2}k_b(x \sin 45^\circ)^2 = \frac{1}{2}k_v x^2, \quad k_v = k_b \sin^2 45^\circ = 1.0 \times 10^7 \text{ N/m}.$$

3. Calculate the natural frequency

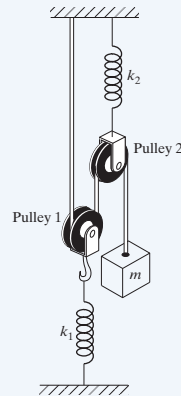
$$m = \frac{2000}{9.81} = 203.874 \text{ kg},$$

$$\omega_n = \sqrt{\frac{10^7}{203.874}} = 221.47 \text{ rad/s}.$$

Thus $f_n = 35.25 \text{ Hz}$. This is the frequency of the specified idealized model. A freely bending real boom requires a different stiffness and motion constraint.

Question 16 | Two spring-supported pulleys**Application**

Determine the natural frequency of the mass m in the system shown. The pulleys and rope are massless, the rope is inextensible, and the pulleys are frictionless. Both springs are linear.



Follow both moving pulley centers when applying the rope constraint.

Solution 16**1. Apply a small additional load**

Apply a downward incremental force P at the mass, measured relative to the loaded equilibrium. Rope tension increases by P throughout the rope. Each pulley therefore experiences an incremental resultant $2P$.

2. Find the pulley movements

Pulley 1 moves upward by $a = 2P/k_1$, and pulley 2 moves downward by $b = 2P/k_2$. Each moving center changes two rope spans. The downward movement of the mass is therefore

$$x = 2a + 2b = 4P \left(\frac{1}{k_1} + \frac{1}{k_2} \right).$$

3. Find the stiffness seen by the mass

$$k_{\text{eq}} = \frac{P}{x} = \frac{k_1 k_2}{4(k_1 + k_2)}.$$

Only m contributes kinetic energy under the stated massless-pulley assumption.

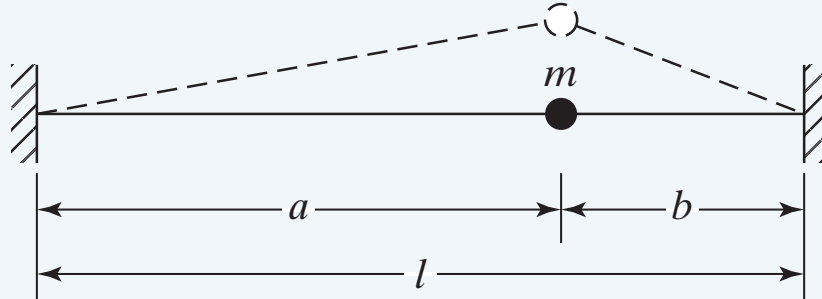
4. Write the vibration equation

$$m\ddot{x} + k_{\text{eq}}x = 0, \quad \omega_n = \sqrt{\frac{k_1 k_2}{4m(k_1 + k_2)}}.$$

If $k_2 \rightarrow \infty$, pulley 2 becomes fixed and $k_{\text{eq}} \rightarrow k_1/4$, consistent with the remaining two-to-one displacement ratio. Rope tension equals mg only in static equilibrium, not throughout the vibration.

Question 17 | A mass on a tensioned cord**Application**

A mass m lies between two fixed cord supports, at distances a and b . The cord tension T is assumed constant during small transverse motion. Neglect cord mass. Derive the equation of motion and natural frequency.



Small transverse displacement from the straight equilibrium line.

Solution 17**1. Resolve the two tensions**

Let y be displacement normal to the equilibrium cord. The restoring component is

$$F_y = -T \frac{y}{\sqrt{a^2 + y^2}} - T \frac{y}{\sqrt{b^2 + y^2}}.$$

The tension acts along each cord segment, toward its fixed support.

2. Linearize for small displacement

For $|y| \ll a, b$, the denominators may be replaced by a and b to first order:

$$F_y \simeq -T \left(\frac{1}{a} + \frac{1}{b} \right) y.$$

Newton's law then gives

$$\boxed{m\ddot{y} + T \left(\frac{1}{a} + \frac{1}{b} \right) y = 0}, \quad \boxed{\omega_n = \sqrt{\frac{T(a+b)}{mab}}}.$$

3. Check the energy and assumptions

The additional cord path length is approximately

$$\Delta L = \frac{y^2}{2a} + \frac{y^2}{2b}.$$

The work against the maintained tension is $U = T\Delta L = \frac{1}{2}k_{eq}y^2$, confirming $k_{eq} = T(1/a + 1/b)$. The model presumes the straight equilibrium line is appropriate; substantial gravity-induced sag or a changing tension requires a different model.

Question 18 | A lifting magnet releases its load**Application**

A 1500 kg lifting magnet holds a 900 kg car. The cable has equivalent axial stiffness 1.75×10^6 N/m. Starting from rest, the magnet releases the car suddenly. Determine the magnet's subsequent vibration and the maximum and minimum cable tensions. Neglect cable mass and damping.

Solution 18**1. Compare the two equilibrium positions**

Take downward displacement z from the unstretched cable position. Before release,

$$z_{\text{old}} = \frac{(1500 + 900)g}{k}.$$

After release, $z_{\text{new}} = 1500g/k$. There is no position jump or velocity impulse, so for $x = z - z_{\text{new}}$,

$$x_0 = \frac{900g}{k} = 0.00504514 \text{ m}, \quad v_0 = 0.$$

2. Find the response of the remaining mass

$$\omega_n = \sqrt{\frac{1.75 \times 10^6}{1500}} = 34.1565 \text{ rad/s},$$

$$x(t) = 0.00504514 \cos(34.1565t) \text{ m}.$$

The full extension is $z(t) = 1500g/k + x(t)$. Initial acceleration is upward because the cable is stretched more than the remaining mass requires.

3. Recover tension from total extension

$$F_c(t) = kz(t) = 1500g + 900g \cos(34.1565t).$$

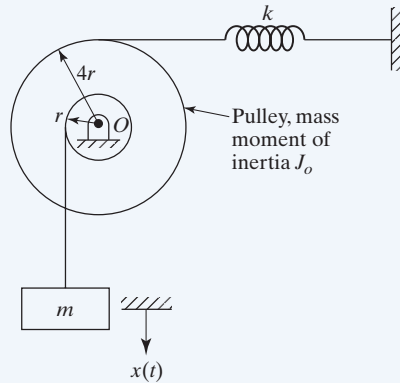
Consequently,

$$F_{c,\text{max}} = 2400g = 23,544 \text{ N}, \quad F_{c,\text{min}} = 600g = 5886 \text{ N}.$$

The minimum is positive, so the cable remains taut and the linear spring model is consistent throughout the motion.

Question 19 | Pulley inertia and two radii**Challenge**

A fixed-axis pulley has mass moment of inertia J_O . A mass m is attached at radius r and a spring k at radius $4r$, as shown. Cords do not slip. Write the force and moment balances and derive the equation using both Newton's law and energy; obtain the natural frequency.



Use downward mass displacement x from equilibrium.

Solution 19**1. Relate the motions**

Choose pulley rotation positive when the mass descends. Then $x = r\theta$, and the incremental spring extension is $u = 4r\theta = 4x$.

2. Newton's law and the pulley moment balance

Let ΔF be the increase in upward tension on the mass. Its incremental balance is $m\ddot{x} = -\Delta F$. About the pulley axis, the mass-side tension drives positive rotation and the spring opposes it:

$$J_O\ddot{\theta} = r\Delta F - (4r)ku = -mr\ddot{x} - 16krx.$$

Use $\ddot{\theta} = \ddot{x}/r$ and divide by r :

$$\left(m + \frac{J_O}{r^2}\right)\ddot{x} + 16kx = 0.$$

3. Check independently with energy

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J_O(\dot{x}/r)^2, \quad U = \frac{1}{2}k(4x)^2.$$

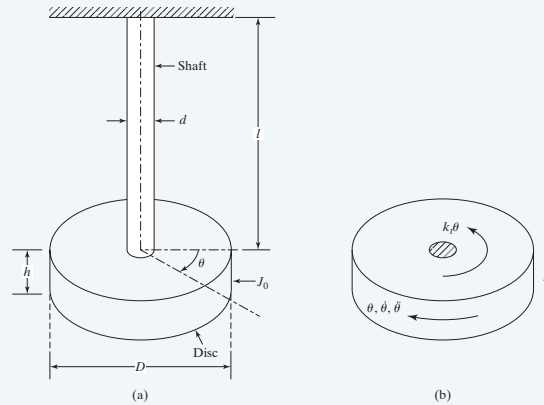
Gravity and static spring preload cancel when expanding about equilibrium. Differentiating $T + U$ gives the same equation, hence

$$\omega_n = \sqrt{\frac{16k}{m + J_O/r^2}}.$$

The square of the radius ratio multiplies stiffness. Pulley inertia contributes J_O/r^2 to the mass associated with coordinate x .

Question 20 | Shaft-and-disk free response**Foundation**

A rigid disk of mass moment of inertia J_O is attached to a massless solid circular shaft of length L , diameter d , and shear modulus G , fixed at its other end. Derive the torsional equation and natural frequency, and obtain the response for initial angle θ_0 and angular velocity $\dot{\theta}_0$.



Shaft-and-disk model and restoring-torque direction.

Solution 20**1. Find the restoring torque**

The shaft's geometric polar area moment is $I_p = \pi d^4/32$. Elastic torsion gives torque magnitude $GI_p\theta/L$, opposing positive twist:

$$k_t = \frac{GI_p}{L}, \quad M_{\text{restoring}} = -k_t\theta.$$

2. Apply angular momentum balance

$$J_O\ddot{\theta} = -k_t\theta, \quad \omega_n = \sqrt{\frac{G\pi d^4}{32LJ_O}}.$$

Here I_p has units m^4 ; J_O has units kg m^2 . They are not interchangeable.

3. Apply the two initial conditions

From $\theta = C \cos \omega_n t + D \sin \omega_n t$, evaluation at $t = 0$ gives $C = \theta_0$ and $D = \dot{\theta}_0/\omega_n$. Thus

$$\theta(t) = \theta_0 \cos \omega_n t + \frac{\dot{\theta}_0}{\omega_n} \sin \omega_n t.$$

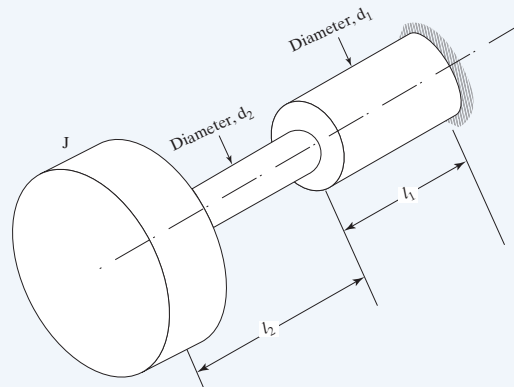
Equivalently, $\theta = \Theta \cos(\omega_n t - \phi_\theta)$, where

$$\Theta = \sqrt{\theta_0^2 + (\dot{\theta}_0/\omega_n)^2}, \quad \phi_\theta = \text{atan2}(\dot{\theta}_0/\omega_n, \theta_0).$$

The corresponding total energy is $\frac{1}{2}J_O\dot{\theta}^2 + \frac{1}{2}k_t\theta^2 = \frac{1}{2}k_t\Theta^2$.

Question 21 | Disk on a stepped shaft**Application**

A heavy disk with mass moment of inertia J is attached to the free end of a stepped circular shaft. The segments have lengths L_1, L_2 and diameters d_1, d_2 , with the same shear modulus G . Neglect shaft inertia and determine the torsional natural frequency.



Two shaft segments connected in series.

Solution 21**1. Relate torque and total twist**

With massless shaft segments, the same torque M_t passes through both segments. Their twists add:

$$\theta = \theta_1 + \theta_2 = M_t \left(\frac{L_1}{GI_{p1}} + \frac{L_2}{GI_{p2}} \right), \quad I_{pi} = \frac{\pi d_i^4}{32}.$$

2. Obtain equivalent torsional stiffness

$$k_t = \frac{M_t}{\theta} = \left(\frac{L_1}{GI_{p1}} + \frac{L_2}{GI_{p2}} \right)^{-1} = \frac{\pi G}{32(L_1/d_1^4 + L_2/d_2^4)}.$$

3. Use the disk inertia

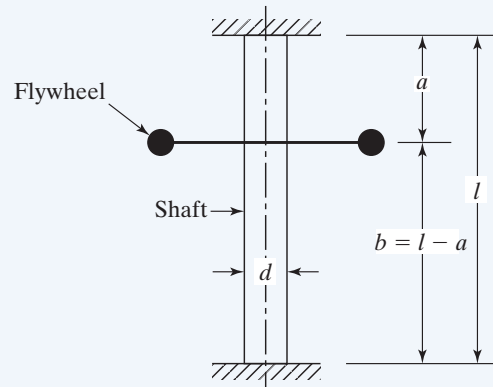
The equation is $J\ddot{\theta} + k_t\theta = 0$, giving

$$\omega_n = \sqrt{\frac{\pi G}{32J(L_1/d_1^4 + L_2/d_2^4)}}, \quad f_n = \frac{\omega_n}{2\pi}.$$

For $d_1 = d_2 = d$, this reduces to the uniform shaft result with length $L_1 + L_2$. The diameter enters to the fourth power, so a short thin segment can dominate total compliance.

Question 22 | Flywheel between two fixed ends**Application**

A flywheel of weight W and radius of gyration r_g is mounted on a uniform circular shaft fixed at both ends. Its distances from the fixed ends are a and $b = L - a$. The shaft diameter is d and shear modulus is G . Neglect shaft inertia and determine the **torsional** natural frequency.



The flywheel twists both shaft segments.

Solution 22**1. Identify the common coordinate**

If the flywheel rotates through θ , both shaft segments have end-twist magnitude θ . Each exerts a restoring torque on the flywheel. Therefore these segments act in parallel, not in series:

$$k_t = \frac{GI_p}{a} + \frac{GI_p}{b} = GI_p \frac{a+b}{ab}, \quad I_p = \frac{\pi d^4}{32}.$$

2. Convert weight to rotational inertia

$$m = \frac{W}{g}, \quad J = mr_g^2 = \frac{Wr_g^2}{g}.$$

3. Write the rotational equation

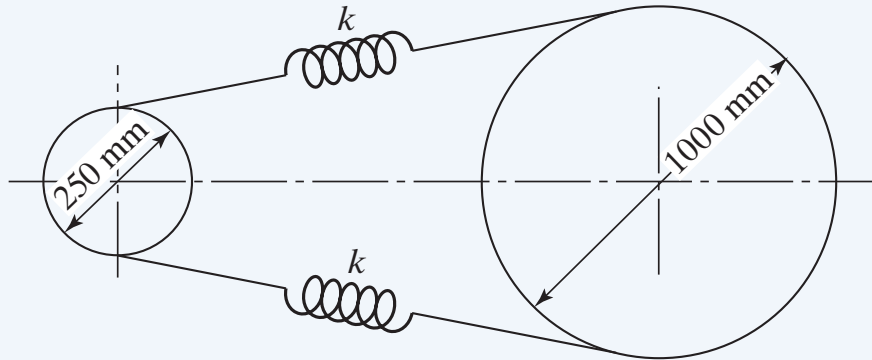
$$J\ddot{\theta} + k_t\theta = 0,$$

$$\omega_n = \sqrt{\frac{G\pi d^4 g L}{32abWr_g^2}}.$$

For a centered flywheel, $a = b = L/2$ and $k_t = 4GI_p/L$. Moving it toward either ideal fixed end increases stiffness because one segment becomes shorter. That limiting trend is meaningful only while the ideal shaft-and-rigid-flywheel assumptions remain appropriate.

Question 23 | Belt stiffness for a torsional frequency**Application**

A 250 mm driving pulley is connected by a belt to a 1000 mm driven pulley of inertia $J = 0.2 \text{ kg m}^2$. Model the two belt spans as springs, each of stiffness k . Find k for a torsional natural frequency of 6 Hz. For this one-coordinate model, take the driving pulley to have zero angular perturbation and assume both belt spans stay taut.



Elastic belt spans; driven pulley radius $R = 0.5 \text{ m}$.

Solution 23**1. Express the belt deformation**

A small angular perturbation θ of the driven pulley changes each span length in magnitude by $R\theta$. One span lengthens and the other shortens. Their incremental restoring torques add.

2. Find the torsional stiffness

The change in strain energy about the preloaded equilibrium is

$$U = \frac{1}{2}k(R\theta)^2 + \frac{1}{2}k(R\theta)^2 = \frac{1}{2}(2kR^2)\theta^2.$$

Thus $k_t = 2kR^2$ and $J\ddot{\theta} + 2kR^2\theta = 0$.

3. Calculate each span stiffness

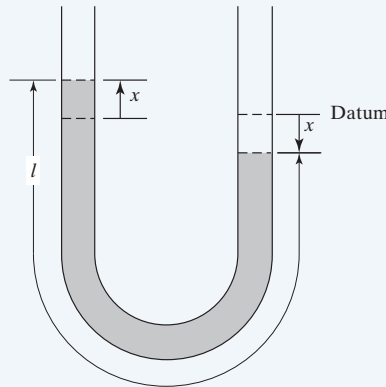
$$\omega_n = 2\pi f_n = 2\pi(6) = 37.6991 \text{ rad/s},$$

$$k = \frac{J\omega_n^2}{2R^2} = \frac{0.2(37.6991)^2}{2(0.5)^2} = 568.49 \text{ N/m}.$$

The smaller diameter does not enter this result because its angular perturbation is constrained to zero. If both pulley rotations were free, a second inertia and an additional coordinate would be needed; the given one-coordinate calculation would no longer apply.

Question 24 | Manometer length for an engine**Application**

A U-tube manometer measures exhaust pressure from a single-cylinder, four-stroke diesel engine running at 600 rpm. Find the minimum total length L of the moving mercury column if its natural frequency must be no greater than the pressure-pulse frequency divided by 3.5. Assume uniform tube area, incompressible liquid, and negligible losses.



A displacement x of either surface creates a level difference $2x$.

Solution 24**1. Use the liquid-column natural frequency**

The entire liquid moves with speed $\dot{x}$, so $T = \frac{1}{2}\rho AL\dot{x}^2$. The level difference produces restoring force $2\rho Agx$, giving $U = \rho Agx^2$. Therefore

$$\omega_n = \sqrt{\frac{2g}{L}}, \quad f_n = \frac{1}{2\pi} \sqrt{\frac{2g}{L}}.$$

2. Convert the engine speed to pulse frequency

A single-cylinder four-stroke engine produces one exhaust event every two revolutions:

$$f_p = \frac{600}{60 \cdot 2} = 5 \text{ Hz}.$$

3. Apply the design inequality

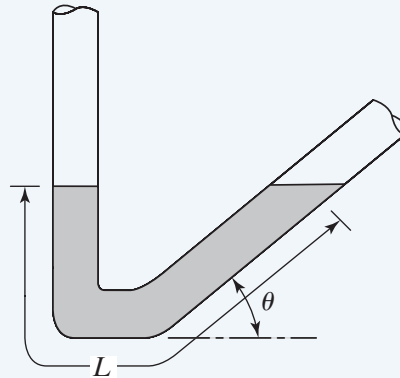
$$f_n \leq \frac{5}{3.5} = 1.42857 \text{ Hz} \implies L \geq \frac{2g}{[2\pi(5/3.5)]^2}.$$

$$\boxed{L_{\min} = 0.24352 \text{ m} \approx 244 \text{ mm}}.$$

L is the total moving liquid length, including the bend and both legs, not the height of one leg. A longer column lowers the natural frequency. The density cancels because it multiplies both inertia and gravitational restoring force. No forced-response or damping calculation is required here.

Question 25 | Inclined manometer**Application**

A uniform-area manometer has one vertical leg and one leg inclined at angle α to the horizontal. The total moving liquid length is L . Derive its natural frequency for small free oscillations, neglecting viscosity and liquid compressibility.



The inclination angle is measured from the horizontal.

Solution 25**1. Define a common displacement coordinate**

Let s be the distance the liquid advances along the inclined leg. Equal tube areas require the vertical surface to descend by the same distance s . The inclined surface rises vertically by $s \sin \alpha$.

2. Calculate the restoring head

The level difference is

$$\Delta h = s + s \sin \alpha = (1 + \sin \alpha)s.$$

The resulting pressure difference is $\rho g \Delta h$. The force opposing the motion is therefore $-\rho A g (1 + \sin \alpha)s$.

3. Include all moving liquid

The liquid mass is $\rho A L$, so

$$\rho A L \ddot{s} + \rho A g (1 + \sin \alpha)s = 0,$$

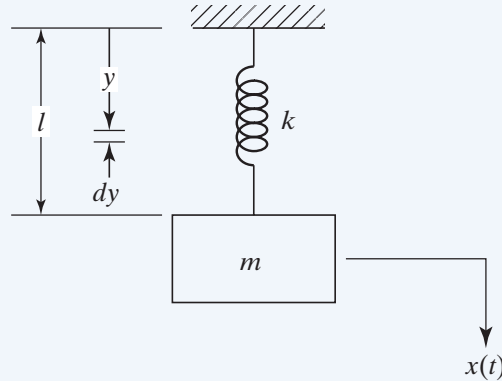
$$\omega_n = \sqrt{\frac{g(1 + \sin \alpha)}{L}}.$$

4. Check limiting cases

At $\alpha = 90^\circ$, both legs are vertical and $\omega_n = \sqrt{2g/L}$. In the horizontal-leg limit, $\alpha = 0$, only the vertical surface changes elevation and $\omega_n = \sqrt{g/L}$. The displacement along an inclined tube must not be confused with its vertical rise.

Question 26 | Including the spring's own mass**Challenge**

A uniform spring of stiffness k , length L , and mass m_s carries an end mass m . Use a displacement field proportional to distance from its fixed end to estimate the natural frequency, including spring inertia.



Displacement is measured about the loaded equilibrium.

Solution 26**1. Describe the assumed motion**

If the end displacement is $x(t)$, a point at coordinate s has

$$u(s, t) = \frac{s}{L}x(t), \quad \dot{u}(s, t) = \frac{s}{L}\dot{x}(t), \quad dm = \frac{m_s}{L} ds.$$

This field satisfies zero displacement at the fixed end and displacement x at the mass.

2. Integrate the distributed kinetic energy

$$T_s = \frac{1}{2} \int_0^L \dot{u}^2 dm = \frac{m_s \dot{x}^2}{2L^3} \int_0^L s^2 ds = \frac{m_s \dot{x}^2}{6}.$$

Thus $T = \frac{1}{2}(m + m_s/3)\dot{x}^2$, while $U = \frac{1}{2}kx^2$ after the static gravitational terms cancel.

3. Equate energy maxima at different times

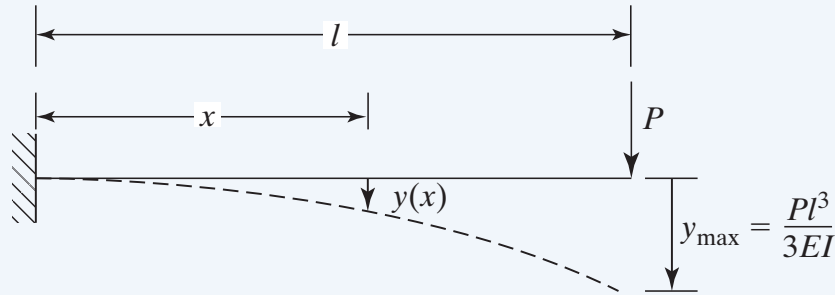
For $x = X \cos \omega_n t$,

$$\frac{1}{2}(m + m_s/3)\omega_n^2 X^2 = \frac{1}{2}kX^2, \quad \omega_n \simeq \sqrt{\frac{k}{m + m_s/3}}.$$

This is an assumed-shape estimate, not an exact distributed-spring solution. Compared with the massless-spring result, the frequency ratio is $\sqrt{m/(m + m_s/3)} < 1$. Not all of the spring moves at the end-mass speed, which explains the factor $1/3$.

Question 27 | Cantilever stiffness and effective mass**Challenge**

A uniform cantilever has length L , bending rigidity EI , total mass m_b , and a point mass M at its tip. Use its static tip-load deflection shape to derive the effective stiffness, effective mass, and Rayleigh estimate of natural frequency. Let the tip displacement be $q(t)$.



Static tip-load shape, normalized by the tip deflection.

Solution 27**1. Normalize the static deflection shape**

For a tip load P , $y(s) = Ps^2(3L - s)/(6EI)$ and $y(L) = PL^3/(3EI)$. Their ratio gives

$$y(s, t) = \psi(s)q(t), \quad \psi(s) = \frac{s^2(3L - s)}{2L^3}.$$

With $r = s/L$, $\psi = (3r^2 - r^3)/2$. It satisfies $\psi(0) = \psi'(0) = 0$ and $\psi(L) = 1$.

2. Calculate the equivalent mass

$$T = \frac{1}{2} \left[M + \frac{m_b}{L} \int_0^L \psi^2(s) ds \right] \dot{q}^2.$$

The dimensionless integral is

$$\int_0^1 \frac{(3r^2 - r^3)^2}{4} dr = \frac{1}{4} \left(\frac{9}{5} - \frac{6}{6} + \frac{1}{7} \right) = \frac{33}{140}.$$

Hence $m_{eq} = M + 33m_b/140$.

3. Calculate stiffness using the same shape

Since $\psi''(s) = 3(L - s)/L^3$,

$$U = \frac{1}{2} EI q^2 \int_0^L (\psi'')^2 ds = \frac{1}{2} EI q^2 \frac{9}{L^6} \frac{L^3}{3} = \frac{1}{2} \frac{3EI}{L^3} q^2.$$

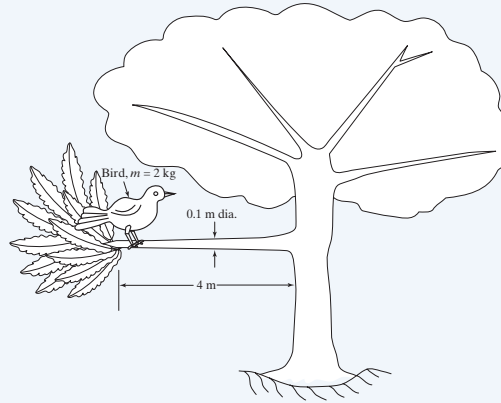
Therefore

$$k_{eq} = \frac{3EI}{L^3}, \quad \omega_n \simeq \sqrt{\frac{3EI/L^3}{M + 33m_b/140}}.$$

The beam factor 33/140 differs from the spring factor 1/3 because their assumed displacement fields differ. This slender-beam estimate neglects shear deformation and rotary inertia.

Question 28 | Bird on a flexible branch**Challenge**

A 2 kg bird sits at the free end of a horizontal branch 4 m long and 0.1 m in diameter. Treat the branch as a uniform circular cantilever with density 700 kg/m^3 and $E = 10 \text{ GPa}$. Form an equation of motion including branch mass and estimate the natural frequency using the static tip-load shape.



Uniform cantilever approximation of the horizontal branch.

Solution 28**1. Evaluate geometric and mass properties**

$$A_b = \frac{\pi(0.1)^2}{4} = 7.85398 \times 10^{-3} \text{ m}^2,$$

$$I = \frac{\pi(0.1)^4}{64} = 4.90874 \times 10^{-6} \text{ m}^4, \quad m_b = \rho A_b L = 21.9911 \text{ kg}.$$

2. Use one consistent assumed shape

For $y(s, t) - y_{\text{static}}(s) = \psi(s)q(t)$ with $\psi = (3r^2 - r^3)/2$,

$$k_{\text{eq}} = \frac{3EI}{L^3} = 2300.97 \text{ N/m}, \quad m_{\text{eq}} = 2 + \frac{33}{140}(21.9911) = 7.18363 \text{ kg}.$$

3. Form the vibration equation and solve for frequency

About the gravity-loaded static equilibrium,

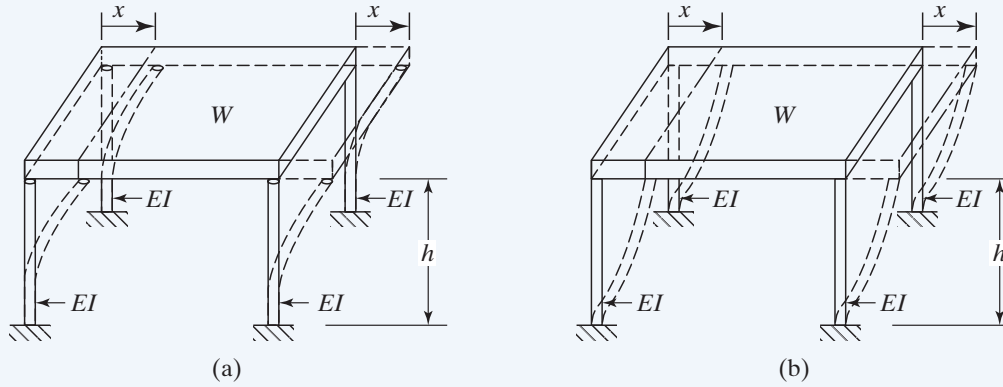
$$7.18363\ddot{q} + 2300.97q = 0,$$

$$\omega_n \simeq 17.8971 \text{ rad/s}, \quad f_n \simeq 2.84842 \text{ Hz}.$$

The branch's distributed weight sets its static sag; its distributed mass contributes the kinetic-energy term. Neglecting that mass would give $\sqrt{2300.97/2} = 33.919 \text{ rad/s}$, substantially overestimating the frequency. The approximation assumes small bending motion about equilibrium and a rigid attachment at the trunk.

Question 29 | Column mass and connection restraint**Challenge**

A rigid floor of weight W is supported by four identical columns, each of height h , weight w , and bending rigidity EI , clamped to the ground. Estimate the horizontal natural frequency including column inertia for (a) pivoted floor connections and (b) connections fixed against rotation. Assume symmetric sway with no floor rotation, and neglect axial-load effects on bending stiffness.



(a) Pivoted upper connections.

(b) Rotation restrained.

Solution 29**1. Define the shared floor coordinate**

Let q be the horizontal floor translation and $r = s/h$. Each column has $y(s, t) = \psi(r)q(t)$. Four stiffnesses and four equivalent column masses add in parallel:

$$m_{\text{eq}} = \frac{W}{g} + 4 \frac{w}{g} \int_0^1 \psi^2 dr, \quad k_{\text{eq}} = 4 \frac{EI}{h^3} \int_0^1 (\psi_{,rr})^2 dr.$$

2. Case (a): pivoted upper connections

The upper end carries shear but no bending moment, so use the cantilever tip-load shape $\psi_a = (3r^2 - r^3)/2$. The previously derived integrals are 33/140 for mass and 3 for stiffness. Thus

$$m_a = \frac{1}{g} \left(W + \frac{33}{35} w \right), \quad k_a = \frac{12EI}{h^3},$$

$$\omega_n^{(a)} \simeq \sqrt{\frac{12EIg}{h^3(W + 33w/35)}}.$$

3. Case (b): zero rotation at both ends

The boundary conditions are $\psi(0) = \psi_{,r}(0) = 0$, $\psi(1) = 1$, and $\psi_{,r}(1) = 0$. A cubic $Ar^2 + Br^3$ gives $A + B = 1$ and $2A + 3B = 0$, so $A = 3$, $B = -2$:

$$\psi_b = 3r^2 - 2r^3.$$

Its mass integral is

$$\int_0^1 \psi_b^2 dr = \int_0^1 (9r^4 - 12r^5 + 4r^6) dr = \frac{9}{5} - 2 + \frac{4}{7} = \frac{13}{35}.$$

Its stiffness integral is

$$\int_0^1 (6 - 12r)^2 dr = 36 \int_0^1 (1 - 4r + 4r^2) dr = 12.$$

Solution 29 (continued)

Hence

$$m_b = \frac{1}{g} \left(W + \frac{52}{35} w \right), \quad k_b = \frac{48EI}{h^3},$$

$$\omega_n^{(b)} \simeq \sqrt{\frac{48EIg}{h^3(W + 52w/35)}}.$$

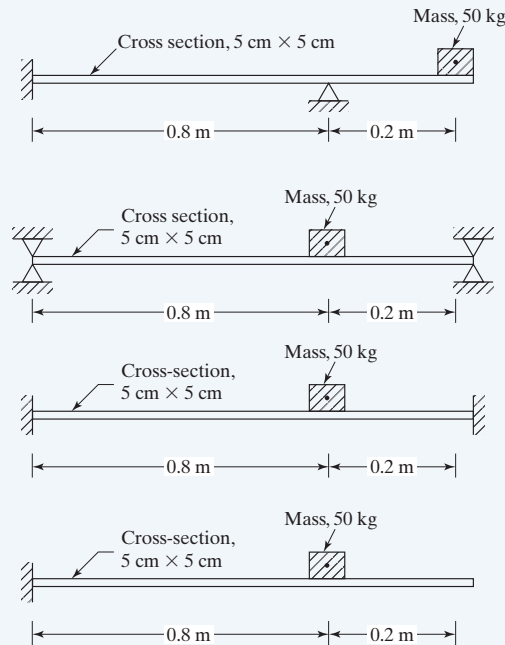
4. Compare the two physical models

$$\frac{\omega_n^{(b)}}{\omega_n^{(a)}} = 2 \sqrt{\frac{W + 33w/35}{W + 52w/35}}.$$

When column mass is negligible, the ratio approaches 2, because restraint multiplies stiffness by four. With column inertia, the shapes and therefore the equivalent masses also differ. Reusing 33/140 for both support cases would be inconsistent.

Question 30 | Four beam-support arrangements**Challenge**

For each arrangement shown, estimate the bending natural frequency using a point mass $M = 50 \text{ kg}$ and a massless steel beam with square cross section $5 \text{ cm} \times 5 \text{ cm}$ and $E = 207 \text{ GPa}$. Number the arrangements (a)–(d) from top to bottom. Use $a = 0.8 \text{ m}$, $b = 0.2 \text{ m}$, and $L = a + b = 1 \text{ m}$.

**Solution 30****1. Choose the coordinate and material properties**

Use vertical displacement at the mass. Apply a force P there to find the compliance $C = \delta/P$; then $k = 1/C$ and $\omega_n = \sqrt{k/M}$. The area moment and bending rigidity are

$$I = \frac{0.05^4}{12} = 5.208333 \times 10^{-7} \text{ m}^4, \quad EI = 107,812.5 \text{ N m}^2.$$

2. Case (a): propped cantilever with an overhang

The prop is at a and the mass is at the free end L . Temporarily remove the prop. A tip force produces deflection at the prop of $Pa^2(3L - a)/(6EI)$. An upward prop force R produces deflection $-Ra^3/(3EI)$ there. Compatibility requires

$$R = \frac{P(3L - a)}{2a}.$$

At the free end, superposition gives

$$\delta = \frac{PL^3}{3EI} - \frac{Ra^2(3L - a)}{6EI}.$$

Substituting R and $L = a + b$ gives

$$C_a = \frac{4L^3 - a(3L - a)^2}{12EI} = \frac{b^2(3a + 4b)}{12EI}.$$

Solution 30 (continued)**3. Case (b): simply supported beam**

The point load is at a from the left and b from the right support. Reactions are Pb/L and Pa/L . The bending moments are $M_1(s) = Pb s/L$ for $0 \leq s \leq a$ and $M_2(s) = Pa(L-s)/L$ for $a \leq s \leq L$. Using strain energy and $\delta = \partial U / \partial P$,

$$C_b = \frac{1}{EI} \left[\int_0^a \left(\frac{bs}{L} \right)^2 ds + \int_a^L \left(\frac{a(L-s)}{L} \right)^2 ds \right] = \frac{a^2 b^2}{3EIL}.$$

4. Case (c): beam clamped at both ends

Let the left end moment and vertical reaction be M_A and R_A . Integrating $EIy'' = M_A + R_A s - P\langle s-a \rangle$ with $y(0) = y'(0) = 0$ gives

$$EIy' = M_A s + \frac{R_A s^2}{2} - \frac{P}{2} \langle s-a \rangle^2,$$

$$EIy = \frac{M_A s^2}{2} + \frac{R_A s^3}{6} - \frac{P}{6} \langle s-a \rangle^3.$$

Here $\langle s-a \rangle = 0$ for $s < a$ and $s-a$ for $s \geq a$. Apply $y(L) = y'(L) = 0$ and solve:

$$M_A = -\frac{Pab^2}{L^2}, \quad R_A = \frac{Pb^2(3a+b)}{L^3}.$$

At the load point, substitution gives $y(a) = -Pa^3b^3/(3EIL^3)$ with this sign convention. Therefore the positive compliance is

$$C_c = \frac{|y(a)|}{P} = \frac{a^3 b^3}{3EIL^3}.$$

5. Case (d): cantilever with the mass at a

The unloaded overhang beyond the mass carries no bending moment. The deflection at the force is that of a cantilever of loaded length a :

$$\delta = \frac{Pa^3}{3EI}, \quad C_d = \frac{a^3}{3EI}.$$

6. Calculate and compare the frequencies

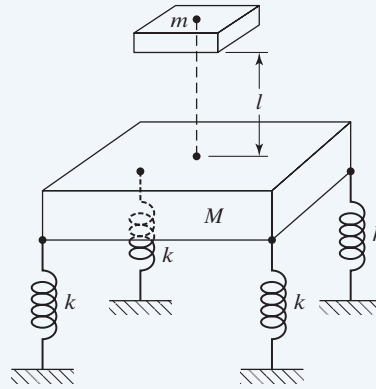
Substitute the dimensions in each compliance and use $k = 1/C$:

Case	k (N/m)	ω_n (rad/s)	f_n (Hz)
(a)	1.01074×10^7	449.61	71.56
(b)	1.26343×10^7	502.68	80.00
(c)	7.89642×10^7	1256.70	200.01
(d)	6.31714×10^5	112.40	17.89

The identical material, cross section, and mass do not imply identical frequencies. Support restraint and the force/measurement location determine the effective stiffness. These answers neglect beam inertia as specified; including it would require a consistent assumed displacement shape for each arrangement.

Question 31 | Falling mass on four elastic supports**Application**

A rigid block of mass M rests on four identical vertical springs, each of stiffness k . A mass m falls through height h onto the center of the block and sticks without rebounding. Neglect spring mass and damping, and assume pure vertical translation with a short impact. Find (a) the natural frequency before impact, (b) the natural frequency afterward, and (c) the complete post-impact response, amplitude, and phase. The block is initially at rest.



The drop height labeled l in the figure is denoted by h here.

Solution 31**1. Combine the four supports and find both frequencies**

All four springs have the same incremental displacement, so their forces add: $K = 4k$. Before impact only M moves; afterward the attached mass is $M + m$. Thus

$$\omega_n^- = \sqrt{\frac{4k}{M}}, \quad \omega_n^+ = \sqrt{\frac{4k}{M + m}}.$$

The added mass decreases the natural frequency. Write $\omega_n = \omega_n^+$ in the following response.

2. Convert the falling motion into an initial velocity

The speed just before contact follows from $mgh = \frac{1}{2}mu_-^2$:

$$u_- = \sqrt{2gh}.$$

Take downward as positive. During a short collision, neglect the finite spring and gravitational impulses compared with the contact impulse. Momentum balance gives

$$mu_- + M(0) = (M + m)v_0, \quad v_0 = \frac{m}{M + m}\sqrt{2gh}.$$

3. Account for the change in static equilibrium

Let z be the downward displacement from the unloaded support position. Before impact $z_- = Mg/K$; the new equilibrium is $z_+ = (M + m)g/K$. Position cannot jump during impact. For $x = z - z_+$,

$$x_0 = z_- - z_+ = -\frac{mg}{K} = -\frac{mg}{4k}.$$

Thus the combined mass starts above its new equilibrium while moving downward.

Solution 31 (continued)**4. Solve the post-impact free vibration**

The equation in z is $(M + m)\ddot{z} + Kz = (M + m)g$. Substitution of $z = z_+ + x$ gives $(M + m)\ddot{x} + Kx = 0$. Hence

$$x(t) = -\frac{mg}{K} \cos \omega_n t + \frac{m\sqrt{2gh}}{(M + m)\omega_n} \sin \omega_n t, \quad t \geq 0.$$

Time is measured from impact. The displacement from the unloaded position is

$$z(t) = \frac{(M + m)g}{K} + x(t).$$

5. Obtain amplitude and phase

For the minus-phase form $x = A \cos(\omega_n t - \phi)$,

$$A = \sqrt{\left(\frac{mg}{K}\right)^2 + \frac{2gh m^2}{K(M + m)}},$$

$$\phi = \text{atan2}\left(\frac{m\sqrt{2gh}}{(M + m)\omega_n}, -\frac{mg}{K}\right).$$

For $h > 0$, the phase is in quadrant II. The maximum downward deflection from the unloaded position is $z_{\max} = (M + m)g/K + A$.

6. Check the collision and response

The kinetic energy lost during sticking is

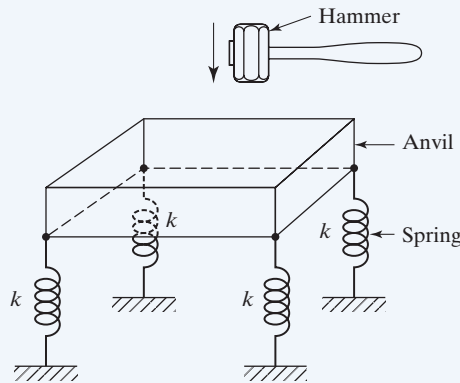
$$\Delta T = \frac{1}{2}mu_-^2 - \frac{1}{2}(M + m)v_0^2 = \frac{1}{2} \frac{mM}{M + m} u_-^2 > 0.$$

Therefore energy conservation must not be applied across this collision. Mechanical energy is conserved during the subsequent ideal undamped motion. Finally, $z(0) = Mg/K$ and $\dot{z}(0) = v_0$, confirming continuity of position and the momentum-derived velocity.

Question 32 | Hammer impact: sticking or separation**Challenge**

A 6 kg sledgehammer strikes a stationary 50 kg anvil at 15 m/s downward. Four springs support the anvil; each has stiffness 17.5 kN/m. Neglect damping and spring mass and assume a short, centered impact with vertical translation only. Determine the anvil response when:

- The hammer remains attached to the anvil after impact (ideal sticking).
- The hammer separates after impact. Express the answer in terms of the coefficient of restitution e , since its value is not specified.



The anvil is initially at its own gravity-loaded equilibrium.

Solution 32**1. Set the stiffness and impact convention**

Take downward as positive and $t = 0$ at impact. The combined stiffness is

$$K = 4(17,500) = 70,000 \text{ N/m.}$$

The incoming momentum is $6(15) = 90 \text{ kg m/s}$. During the short collision, external spring and gravitational impulses are neglected.

2. Part (a): find the common velocity and new equilibrium

Momentum balance gives

$$v_0 = \frac{90}{50 + 6} = 1.607143 \text{ m/s.}$$

The equilibrium shifts downward by $6g/K$. Measuring x_a from this new equilibrium gives

$$x_{a0} = -\frac{6(9.81)}{70,000} = -0.000840857 \text{ m,} \quad \omega_n^{(a)} = \sqrt{\frac{70,000}{56}} = 35.35534 \text{ rad/s.}$$

3. Part (a): write the complete response

Using $x_a = x_{a0} \cos \omega_n^{(a)} t + (v_0/\omega_n^{(a)}) \sin \omega_n^{(a)} t$,

$$x_a(t) = -0.000840857 \cos(35.35534t) + 0.04545686 \sin(35.35534t) \text{ m.}$$

The equivalent amplitude and phase are

$$A_a = \sqrt{(-0.000840857)^2 + (0.04545686)^2} = 0.04546464 \text{ m,}$$

Solution 32 (continued)

$$\phi_a = \text{atan2}(0.04545686, -0.000840857) = 1.589292 \text{ rad.}$$

Thus $x_a = 0.04546464 \cos(35.35534t - 1.589292) \text{ m}$.

To recover displacement from the unloaded spring position, add the new static deflection:

$$z_a(t) = \frac{56g}{K} + x_a(t) = 0.007848 + x_a(t) \text{ m.}$$

At $t = 0$, $z_a = 50g/K = 0.00700714 \text{ m}$, exactly the pre-impact position. The initial acceleration is $-\omega_n^{(a)2}x_{a0} = 1.05107 \text{ m/s}^2$ downward, consistent with the added weight.

4. Part (b): use momentum and restitution together

Let u_h and u_A be the hammer and anvil velocities immediately after impact. Momentum supplies one equation:

$$6u_h + 50u_A = 90.$$

Restitution relates separation speed to approach speed:

$$u_A - u_h = 15e.$$

Substitute $u_h = u_A - 15e$ into the first equation:

$$56u_A = 90(1 + e), \quad \boxed{u_A = 1.607143(1 + e) \text{ m/s}},$$

$$u_h = \frac{15(6 - 50e)}{56} \text{ m/s.}$$

Separation does not necessarily mean the hammer moves upward. Upward rebound requires $u_h < 0$, or $e > 6/50 = 0.12$; for smaller positive e both may initially move downward while separating.

5. Part (b): solve the anvil-only free response

After separation only the anvil mass remains on the springs. Its static equilibrium is unchanged, so $x_{b0} = 0$, $\dot{x}_{b0} = u_A$, and

$$\omega_n^{(b)} = \sqrt{\frac{70,000}{50}} = 37.41657 \text{ rad/s.}$$

Therefore

$$\boxed{x_b(t) = 0.04295270(1 + e) \sin(37.41657t) \text{ m}},$$

or, in the cosine's minus-phase form,

$$x_b(t) = 0.04295270(1 + e) \cos(37.41657t - \pi/2) \text{ m.}$$

The displacement from the unloaded spring position is $z_b(t) = 50g/K + x_b(t)$. This free-response expression applies while the hammer and anvil remain separated; any later collision requires a new impact calculation.

6. Interpret the missing information

For an ordinary passive separating impact, $0 < e \leq 1$. The amplitude is $A_b = 42.95270(1 + e) \text{ mm}$. The data do not determine a unique numerical amplitude without e or an equivalent measured post-impact velocity. Simply removing the hammer from the vibrating mass while retaining the sticking velocity silently imposes $e = 0$ and does not describe a general rebound.